

How Institutional Trust Shapes User Trust in Domestic and Global AI Models: Evidence from the Chinese Context

Introduction

The rapid proliferation of large language models (LLMs) has transformed artificial intelligence from a specialized tool into a primary infrastructure for everyday communication, information seeking, and decision-making. Prominent examples such as ChatGPT and DeepSeek have quickly entered public discourse, sparking renewed attention to how trust in AI is socially and culturally constructed.

In this study, we define AI trust as a multidimensional construct comprising cognitive trust and affective trust. The former refers to rational assessment of a system's competence and reliability, while the latter refers to an emotional bond and perceived benevolence of the technology (McAllister, 1995). Current scholarship in human-machine communication (HMC) has predominantly focused on techno-centric factors, such as spanning system performance, interface design (e.g., anthropomorphism), and transparency mechanisms (Hancock et al., 2011). While these studies offer valuable insights into the psychological mechanics of trust, they often treat the user and the machine as analytically isolated from broader institutional contexts, overlooking the macro forces that govern technology adoption. Recent comparative studies have begun to acknowledge that domestic AI systems often evoke higher trust due to linguistic alignment and cultural resonance (Liu & Zhao, 2023). However, a significant theoretical gap persists. Existing literature has yet to systematically examine how institutional trust, an individual's confidence in overarching governance and regulatory systems, moderates the relationship between an AI model's origin and a user's multidimensional trust. Specifically, we lack a clear understanding of how institutional credibility is transferred into technological credibility, and how this process varies across

different user groups within the same national population.

While DeepSeek and ChatGPT differ in technological capabilities, their trust-building mechanisms are shaped by an interplay of functional performance and institutional legitimacy (Fosch-Villaronga & Millard, 2019). Therefore, we introduce the term Institutional Prism to describe the moderating role of macro-level governance in shaping micro-level technological perceptions. Drawing on Pierre Bourdieu's (2014) concept of field refraction, we suggest that the national institutional environment acts as a lens that bends the perception of AI attributes to align with domestic social orders and political legitimacies.

To address these gaps, this study integrates McAllister's (1995) Cognitive-Affective Trust Theory (CATT) with the Institutional Prism framework. The aim of this paper is to investigate how the origin of an AI model (domestic vs. global) influences cognitive and affective trust, and to demonstrate how institutional trust serves as the refraction mechanism shaping these evaluations. By comparing DeepSeek and ChatGPT in the Chinese context, we argue that institutional trust functions as a vital social resource that channels state legitimacy into a relative trust advantage for domestic technologies. Furthermore, we reveal how structural access frictions and institutional signals create divergent pathways for AI adoption, potentially leading to imbalanced epistemic ecosystems.

This research utilizes survey data from 405 Chinese users to empirically test the moderating effects of institutional trust. The paper is structured as follows: first, we develop the Institutional Prism framework and formulate hypotheses based on CATT. Second, we describe the methodology and present the results of our comparative analysis. Finally, we

discuss the theoretical implications of our findings for media sociology and offer practical insights for inclusive AI governance and global technological equity.

Literature Review and Hypotheses

Paradigms of Human-AI Trust from Technical Rationality to Social Agency

Trust, as a fundamental psychological foundation in human–machine collaboration, determines social acceptance, sustained use, and the stability of interactions between users and AI systems (Riegelsberger et al., 2005). In intelligent communication environments, the construction of trust is no longer a purely rational evaluation process but a comprehensive psychological mechanism integrating technological features, social contexts, and cultural affiliation. With the rapid development of computer technology, the study of trust has extended into non-human domains, particularly in the context of human–machine interaction (Y.-K. Lee et al., 2014).

The conceptualization of trust in human-machine communication (HMC) has traditionally been dominated by two competing paradigms. The first is the techno-centric paradigm, which can be specifically categorized into four categories of system-level trust antecedents: effectiveness, transparency, explanation, and human oversight (Aoki, 2021; Grimmelikhuijsen, 2022), heavily informed by information systems research, including but not limited to Technology Acceptance Model (TAM). It assumes that people trust AI when it performs well and distrust it when it does not (König et al., 2022). This perspective approaches user trust as a calculative, rational assessment of an AI system’s functional attributes, such as accuracy, computational reliability, and transparency. Under a techno-

rational view, users may attribute higher cognitive trust to global LLMs because of their perceived scale, sophistication, and performance consistency. Yet cross-national evidence suggests that such capability-based judgments do not fully determine trust (Gillespie et al., 2023).

Conversely, the socio-relational paradigm, rooted in the "Computers Are Social Actors" (CASA) framework, which shifts trust formation from performance-based evaluation to socially embedded perception, posits that as AI systems exhibit increasing autonomy and natural language proficiency, they are evaluated not merely as tools, but as social agents (Nass & Moon, 2000). Scholars emphasize that trust is fostered through anthropomorphic cues, social presence, and cultural alignment. From this viewpoint, domestic AI models possess an inherent advantage (Sun et al., 2024). By leveraging localized linguistic nuances, culturally specific idioms, and shared societal values, domestic systems may achieve a form of epistemic alignment, defined as the perceived congruence between a model's knowledge representation and a user's cultural-cognitive schema, transforming the AI from a foreign utility into an empathetic partner.

While both paradigms offer critical insights, they share a significant theoretical blind spot: they conceptualize human-AI trust within a insufficiently situated within broader socio-political structures. By treating trust as either a purely algorithmic calculation or an intimate interpersonal simulation, existing literature largely fails to account for the macro-structural forces that govern technology, as suggested in broader media governance scholarship (e.g., Flew, 2024). AI systems do not operate independently; they are heavily embedded within national regulatory frameworks, data sovereignty debates, and geopolitical contexts. Recent

research increasingly shows that trust in AI is shaped not only by system features but also by trust in governments, firms, and governance institutions (Amir et al., 2025). Therefore, evaluating user trust in domestic versus global AI requires a multidimensional framework that bridges individual psychological assessment with macro-institutional realities.

The Cognitive-Affective Framework

To capture the complexity of AI evaluation, this study adopts the Cognitive-Affective Trust Theory (CATT). Originally proposed by McAllister (1995) and increasingly applied to HMC, CATT posits that trust is not a unidimensional construct but a bipartite psychological structure comprising cognitive and affective dimensions (Benbasat & Wang, 2005).

Cognitive trust refers to a rational belief in the capability, transparency, and reliability of a trustee (Schoorman et al., 2007). In the context of generative AI, users face high "algorithmic uncertainty" (Kizilcec, 2016). When evaluating cognitive trust, users may face a dilemma: they might recognize the global stability and vast training data of models like ChatGPT (Watson et al., 2023), thereby attributing high computational competence to it. Alternatively, they might perceive domestic models like DeepSeek as possessing higher task-specific reliability and compliance within the local context (Shen et al., 2026). Given the conflicting evidence regarding whether global computational power or local contextual accuracy drives cognitive trust, we propose the following research question:

RQ1: Is there a significant difference in Chinese users' cognitive trust between the domestic AI model (DeepSeek) and the global AI model (ChatGPT)?

Affective trust, distinct from rational calculation, centers on emotional bonds, empathy, and a

sense of belonging formed through interaction (McAllister, 1995; Komiak & Benbasat, 2006). In intelligent communication contexts, affective trust is highly contingent upon cultural alignment. Domestic models are specifically optimized for local cognitive patterns, utilizing polite expressions, cultural metaphors, and familiar narrative structures that enhance a user's sense of social presence (Kim et al., 2012). International models may face an affective disadvantage when users perceive weaker cultural alignment or institutional embeddedness, though this disadvantage can be reduced through localization (Shen et al., 2026). Therefore, we hypothesize:

H1: Chinese users will demonstrate higher affective trust in the domestic AI model (DeepSeek) than in the global AI model (ChatGPT).

The Institutional Prism: Trust Transfer and Refraction Mechanisms

The CATT framework suggests that user evaluations are not formed in a cognitive vacuum; rather, they are profoundly embedded in a broader social architecture (Granovetter, 1985).

While cognitive and affective dimensions explain the internal structure of trust, the direction and intensity of these evaluations are inevitably shaped by the macro-structural environment, specifically the level of institutional trust. Institutional trust, defined as an individual's confidence in the integrity, competence, and regulatory capacity of overarching governance systems (Zucker, 1986), provides the essential "background certainty" that allows users to navigate the inherent "black-box" uncertainty of AI algorithms.

Drawing on Trust Transfer Theory (McKnight et al., 2011), we argue that in data-sensitive domains like AI, trust primarily flows from macro-level institutions to micro-level

technological artifacts (Y.-F. Wang et al., 2024; Kostka et al., 2021), albeit with potential feedback loops. In digital government research, trust in government has long been shown to support trust in e-services and adoption intentions (Horsburgh et al., 2011). To theorize this cross-level interaction, we propose the "Institutional Prism" framework, drawing inspiration from Pierre Bourdieu's concept (Bourdieu, 1990; Hilgers & Mangez, 2014) of field refraction. In Bourdieu's sociology, a field, such as the national technological or communicative field, does not merely reflect external pressures or global forces like a mirror. Instead, it refracts them according to its own internal logic. Just as a physical prism bends light based on its refractive index, the national institutional environment acts as a moderating structure that bends the perception of global technological artifacts to align with domestic social orders.

Unlike blocking information, the institutional prism actively reshapes how an AI's origin is interpreted. For users with high institutional trust, the legitimacy of state governance and domestic regulatory frameworks is successfully transferred to domestic systems (Pande et al., 2025). This institutional endorsement may function as a trust premium, especially when users perceive domestic governance institutions as competent and protective (Y.-F. Wang et al., 2024). Cognitively, it compensates for potential technological gaps, regarding the domestic model as safer, more compliant, and ultimately more reliable. Affectively, the domestic model is refracted as a symbolic extension of national identity and collective security (Kao & Sapp, 2022). In this refractive process, technical attributes are not perceived in their raw form but translated into signals of national security, cultural compliance, and institutional reliability.

Conversely, for users with low institutional trust, this transfer mechanism becomes less

effective (Song et al., 2025). In the absence of institutional endorsement, these users are more likely to retreat to a techno-rational evaluation, prioritizing the perceived independence and objectivity of global models. The refraction shifts toward the perceived independence and techno-neutrality of global models like ChatGPT. In this sense, the Institutional Prism explains why the same global technological force, generative AI, is perceived as a liberating tool by some and a limited utility by others within the same national population. The prism demonstrates that trust is never a direct reflection of algorithmic performance, but a refracted product of the institutional field's internal logic.

Structural Friction as an Institutional Signal

In specific national contexts like China, the institutional prism is further materialized through structural access frictions, such as internet gateways and regulatory restrictions on global AI models. Previous studies often treat such barriers merely as prerequisites for use. However, from a communication policy perspective, it may function as a powerful institutional signal (Schiff et al., 2023).

Li's parallel surveys in China and the United States show that respondents trust their own government most and their geopolitical rival least in the domain of global AI governance (Li, 2025). By imposing boundaries on global models, the institutional environment inherently "others" these foreign technologies, signaling potential risks regarding data sovereignty and informational safety. For high-trust individuals, navigating these frictions reinforces their alignment with the domestic system, deepening their uncertainty toward global models and amplifying their reliance on domestic alternatives (Shen et al., 2026). For low-trust

individuals, however, these same structural frictions may create an alternative evaluative pathway, where restriction may paradoxically enhance the symbolic appeal of global models because of its perceived distance from domestic regulation.

Therefore, institutional trust serves as the critical moderator that dictates how users navigate this complex sociotechnical landscape. Based on the mechanism of the institutional prism, we propose the following hypotheses regarding its moderating effects:

H2: Institutional trust moderates the relationship between AI model type and cognitive trust. Specifically, higher institutional trust reduces the negative predictive effect of DeepSeek on cognitive trust relative to ChatGPT.

H3: Institutional trust moderates the relationship between AI model type and affective trust. Specifically, higher institutional trust amplifies the positive predictive effect of DeepSeek on affective trust relative to ChatGPT.

The proposed theoretical model is illustrated in Figure 1 The theoretical model diagram.

Methodology

Sample and Procedure

This study was conducted from February to April 2025, with data collected via the Credamo platform, which handled questionnaire design, distribution, and collection. A total of 476 questionnaires were collected. After removing invalid responses that failed the attention check, 405 valid samples remained. Since the study includes 13 observation variables, the sample size is adequate for regression-based analyses.

This study focuses on users who are able or consciously evaluate between the two models.

Participants had to meet the following criteria: (1) be aged 18 or older, (2) reside in mainland China, and (3) have used at least one domestic AI product (e.g., DeepSeek) and one international AI product (e.g., ChatGPT) in the past three months, allowing for valid trust comparisons across models.

Descriptive statistics of the 405 valid samples indicate a concentrated age distribution, with an average age of 24.12 years ($SD = 7.61$) and a range from 18 to 59 years. Among participants, 45.9% were aged 18–20, and 29.6% were 21–30. Regarding education, 44.9% had a high school diploma or lower, 48.1% had a college or undergraduate degree, and 6.9% held a master's or doctoral degree, indicating a relatively high educational level. Regarding occupation, 64.7% were students, comprising the majority of the sample. The remaining respondents were primarily from education/research (8.1%), internet/IT (4.2%), finance/law (3.5%), and other sectors.

While the sample is skewed toward younger and student populations, this group represents a key demographic in AI adoption and intensive usage contexts. Moreover, students generally show higher usage frequency, greater willingness to adopt technology, and heightened sensitivity to emerging AI technologies, reinforcing the representativeness and suitability of the sample. The study was approved by the author's university ethics committee with strict adherence to ethical research standards maintained.

Variable Measurement

The primary variables in this study were AI model type (DeepSeek vs. ChatGPT), cognitive

trust, affective trust, and institutional trust, all measured on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree).

Cognitive Trust: This variable was assessed using an adapted version of McAllister's (1995) cognitive trust subscale, modified for AI system contexts. Sample items included: "DeepSeek demonstrates high professionalism and reliability, effectively completing tasks," "Based on DeepSeek's past performance, I believe it is capable of handling complex tasks," and "In actual use, DeepSeek operates accurately and reliably, with minimal technical issues." The scale showed good reliability (Cronbach's $\alpha = 0.789$).

Affective Trust: Affective trust was assessed using an adapted version of McAllister's (1995) affective trust subscale. Example items included: "I am willing to share my work progress, emotional state, and creative ideas with DeepSeek," "When I encounter difficulties, I trust DeepSeek to understand my needs and provide support," and "When submitting feedback or encountering issues, DeepSeek offers empathetic and specific responses." This scale also showed good reliability (Cronbach's $\alpha = 0.728$).

Institutional Trust: This variable reflects the public's trust in government and its actions, specifically concerning AI development. The scale, adapted from Sun et al. (2007), assessed public trust in government institutions and AI-related policies. Example items included: "The government can mobilize resources to address AI development challenges," "AI governance policies are genuinely implemented, not merely for show," and "Government officials responsible for AI governance are trustworthy." This scale demonstrated good reliability (Cronbach's $\alpha = 0.838$).

Control Variables: Demographic variables, including age, gender, education level, and occupation, were controlled in regression and moderation models to account for potential confounding effects.

Analytical Methods

This study employed SPSS software for data analysis and hypothesis testing to ensure the rigor and reliability of the results. The data analysis process comprised descriptive statistics, paired-sample t-tests, and moderation effect analysis. The specific steps are as follows: First, to gain a comprehensive understanding of the sample, descriptive statistical analysis was conducted on the key variables. Descriptive analysis involved calculating the mean, standard deviation, minimum, and maximum values of the variables, providing a preliminary summary of the distribution characteristics and trends of each observed variable. This analysis reveals the basic features of the data and provides a foundation for subsequent analyses. Second, given that different AI models may influence users' cognitive and affective trust to varying degrees, paired-sample t-tests were conducted to compare user scores between DeepSeek and ChatGPT on both dimensions. Finally, moderation analysis was conducted to examine whether institutional trust moderates the relationship between AI model type and user trust. The PROCESS macro developed by Hayes (Model 1) was used for moderation analysis (Hayes, 2017), with institutional trust specified as the moderator, to assess its effect on the relationship between AI model type and both cognitive and affective trust. In this analysis, the presence and strength of the moderation effect were determined by testing the significance of the interaction terms. Additionally, simple slope analysis and plotting moderation effect graphs were used to visually illustrate how institutional trust moderates the

relationship between AI type and user trust at different levels.

Common Method Bias

As this was a cross-sectional study, data were collected simultaneously, which may introduce common method bias (CMB) and potentially affect the accuracy of the results. To minimize this risk, several measures were adopted during the study's design and implementation phases. First, anonymous questionnaires were administered to ensure respondent privacy and reduce social desirability bias. Second, the questionnaire was designed to avoid vague or double-barreled items, promoting clarity and precision. Different measurement scales were employed for different latent variables to minimize systematic error arising from single-source measurement tools. Additionally, during data analysis, Harman's Single Factor Test was performed in SPSS to assess potential common method bias (Podsakoff et al., 2003). The results indicated that the variance explained by the first unrotated factor was 32%, below the 40% threshold, suggesting that common method bias was limited.

Results and Findings

Descriptive Statistics and Correlation Analysis of Key Variables

Preliminary examination of the correlation coefficients between variables showed that all correlations were below 0.7, effectively mitigating potential multicollinearity issues.

Descriptive statistics, including means and standard deviations, are presented in Table 1 Pearson Correlation Matrix of Key Variables. To further examine relationships among the variables, the Pearson correlation matrix for the key variables is presented. The results

indicate significant positive correlations among AI model type, affective trust, and cognitive trust, with institutional trust demonstrating strong associations with both affective and cognitive trust.

Paired-Sample T-Test

To explore the differences in users' trust levels under different AI models (DeepSeek and ChatGPT), this study conducted a paired-sample t-test on the scores of the same participants across two dimensions: cognitive trust and affective trust.

In terms of cognitive trust, the mean cognitive trust score of DeepSeek was 3.58 (SD = 0.59), while that of ChatGPT was 3.51 (SD = 0.66). The paired-sample t-test results indicated a significant difference in cognitive trust between the two models ($t = 2.474$, $p < 0.05$), showing that Chinese users had higher cognitive trust in DeepSeek than in ChatGPT. This result suggests that, although DeepSeek is a domesticized AI model with a relatively short development history, its optimization and adaptation to the Chinese context may have enhanced users' evaluations of its cognitive ability and reliability.

While in terms of affective trust, the mean affective trust score of DeepSeek was 3.44 (SD = 0.73), while that of ChatGPT was 3.18 (SD = 0.63). The paired-sample t-test also revealed a significant difference ($t = 6.697$, $p < 0.001$), further showing that Chinese users had a higher level of affective trust in DeepSeek than in ChatGPT. This result suggests that DeepSeek may better meet users' expectations in emotional interaction, cultural adaptation, and domesticized experience optimization, thereby fostering higher affective trust.

Consequently, users exhibited higher trust in DeepSeek across both cognitive and affective

dimensions, with a stronger preference for affective trust. This finding provides empirical support for understanding similarities and differences in the trust-building mechanisms of different AI models.

Moderating Effect of Institutional Trust

To examine whether institutional trust moderates the relationship between AI type and user trust, the PROCESS macro (Model 1) developed by Hayes (2017) was employed. AI type (DeepSeek vs. ChatGPT) served as the independent variable, institutional trust as the moderator, and cognitive trust (H2) and affective trust (H3) as the dependent variables, to examine the significance of the interaction effect. A total of 5000 bootstrap samples were generated, with a 95% confidence interval. Additionally, simple slope analysis was conducted on significant interaction terms to illustrate how AI type affects user trust at different levels of institutional trust.

This study analyzed the relationship between AI type and cognitive trust and introduced institutional trust as a moderating variable. Results indicated that the interaction term (AI type $\times$ institutional trust) significantly moderated cognitive trust ($p < 0.05$). Specifically, at a low level of institutional trust ($M - 1$ SD), the difference in cognitive trust between DeepSeek and ChatGPT was minimal ($M_{\text{DeepSeek}} = 3.36$, $M_{\text{ChatGPT}} = 3.38$), whereas at a high level of institutional trust ($M + 1$ SD), the difference increased ($M_{\text{DeepSeek}} = 3.81$, $M_{\text{ChatGPT}} = 3.62$). These results suggest institutional trust positively moderates the relationship between AI type and cognitive trust: as institutional trust increases, DeepSeek's cognitive trust advantage becomes more pronounced. This finding supports H2.

It also analyzed the moderating role of institutional trust between AI type and affective trust. The interaction term between AI type and institutional trust also significantly moderates affective trust ($p < 0.001$). Specifically, at a low level of institutional trust, the difference in affective trust between DeepSeek and ChatGPT was minimal ($M_{\text{DeepSeek}} = 3.20$, $M_{\text{ChatGPT}} = 3.11$), whereas at a high level of institutional trust, DeepSeek's affective trust score increased substantially ($M_{\text{DeepSeek}} = 3.68$, $M_{\text{ChatGPT}} = 3.25$). This result indicates that institutional trust positively moderates the relationship between AI type and affective trust. In contexts of high institutional trust, DeepSeek is more likely to elicit greater affective trust and emotional endorsement from users. This finding supports H3.

These findings indicate that institutional trust, as an environmental factor, plays a critical moderating role between AI type and user trust. High institutional trust not only amplifies users' cognitive trust advantage in DeepSeek but also strengthens affective trust and users' sense of belonging. Figure 2 The schematic diagram of the moderating effect schematically illustrates the dynamic trend of trust differences between AI types across levels of institutional trust.

Discussion

This study employed the Cognitive-Affective Trust Theory and the Institutional Prism framework to examine the trust mechanisms underlying domestic and global large language models. The empirical results suggest that trust is not merely a technical response to algorithmic output. It is a socially embedded process where institutional legitimacy significantly affects the perception of technological competence and emotional resonance.

This aligns with recent evidence showing that trust in AI is shaped not only by system features but also by trust in governments, firms, and governance institutions (Amir et al., 2025).

Institutional Compatibility: Redefining Intelligence

A central finding of this research is that cognitive trust in DeepSeek is strengthened by institutional trust. This suggests that in the era of artificial intelligence, the traditional definition of competence, typically centered on technical accuracy and computational power, is undergoing institutionally impacted. For users with high institutional trust, the domestic model is perceived as more capable because it aligns with the regulatory boundaries and social norms of the local environment. This alignment grants the algorithm a form of cognitive legitimacy that transcends raw performance metrics. For users with high institutional trust, an AI model may be perceived as more capable when it is seen as operating appropriately within familiar regulatory and normative boundaries (Pande et al., 2025).

Divergent Pathways and Contextualized Knowledge Ecosystems

The data reveals that different groups have distinct trust evaluation paths when dealing with global and domestic technologies. This reflects that in the context of digital governance; society is forming a contextualized knowledge ecosystem. For high-trust individuals, the institutional prism functions as an integrative force, merging national identity and technical evaluation into a singular, self-reinforcing knowledge system (Sun et al., 2024). However, for those with lower institutional trust, this integration fails, prompting a search for external epistemic authority in global models. While the refraction of global technology through a

national prism is a necessary part of technological localization, we must remain cautious of the potential for cognitive closure. If the institutional prism shifts from a mechanism of healthy refraction to informational insulation, it may have implications for epistemic fragmentation. Maintaining a balance between institutional alignment and epistemic diversity is therefore essential for the long-term health of the digital ecosystem.

The Paradox of Structural Frictions

This study demonstrates that structural frictions in technology access (such as access barriers, etc.) play a complex role in users' minds. We should not merely view them as physical obstacles but rather consider them as a kind of psychological heuristic (Aoki, 2021), producing a dialectical effect on user perception. For users aligned with the domestic institutional discourse, these frictions are interpreted as protective signals that enhance the perceived safety and cultural independence of the domestic AI ecosystem. Conversely, for users with lower institutional trust, the barriers lead to an exaggeration of the global model. Because global AI is perceived as being outside the reach of local institutional constraints, it is often imbued with a sense of absolute objectivity or purity. This can be problematic, as it may cause users to overlook the inherent biases and ideological frameworks that also govern global AI systems.

Adjustable Trust and User Agency

Finally, the results indicate that users are not passive subjects of institutional refraction but active agents who perform strategic trust allocation. Individuals appear to manage their trust resources based on the specific requirements of a given task and the perceived institutional

risks involved. The findings suggest that users may allocate trust strategically rather than uniformly, potentially favoring different models under different perceived institutional and communicative demands. This adjustable trust highlights the sophistication of modern AI users who navigate the institutional prism by utilizing the strengths of different systems to hedge against the perceived limitations of each. This nuanced distribution of trust underscores the complexity of human-machine communication in a globalized yet institutionally partitioned world.

Theoretical Contributions

This study examines how AI model type influences user trust, with particular emphasis on differences between cognitive and affective trust, and the moderating role of institutional trust. The findings not only validate assumptions of existing trust theories but also extend the theoretical framework of Human-Machine Communication (HMC) and sociotechnical systems.

First, this study suggests a more contextualized approach to HMC by incorporating the concept of field refraction theory into the study of algorithmic trust (Hilgers & Mangez, 2014; Couldry, 2003). While traditional research often evaluates user trust as a direct, dyadic response to system capabilities or interface design (Hancock et al., 2011), our findings indicate that this process is mediated by the macro-structural environment. By theorizing institutional refraction as a critical layer between model features and user trust judgments, this study offers a framework for understanding why global technological artifacts are locally reinterpreted. Drawing on Bourdieu's view of field dynamics, we show that generative AI

does not enter a socially neutral vacuum; instead, its perception is refracted by the internal logic of the national technological field. This perspective provides evidence that the origin of an AI model functions as a significant socio-technical signal, moving beyond the techno-centric focus of earlier models.

Second, the study expands the Cognitive-Affective Trust Theory (CATT) by conceptualizing a macro-to-micro trust transfer mechanism (Zucker, 1986). While previous literature relying on McAllister (1995) or Mayer et al. (1995) effectively distinguished the rational and emotional dimensions of trust, it largely confined these dimensions to interpersonal or organizational boundaries. The finding illustrates that cognitive and affective trust are not formed in isolation from broader governance beliefs. Instead, macro-level institutional trust can shape how users interpret competence, safety, and relational alignment in AI systems, consistent with recent evidence on trust transfer in AI-enabled public services (Song et al., 2025). By showing that macro-level confidence in state governance can be systematically transferred into micro-level emotional attachment and perceived competence of domestic AI products, this study provides a novel, ecologically valid theoretical model for sociotechnical trust.

Finally, this research contributes to the global communication discourse by conducting a comparative analysis between a domestic model (DeepSeek) and a global model (ChatGPT). The findings demonstrate that AI systems are encountered by users not as culturally empty computational engines, but as institutionally and nationally situated communicative artifacts (Flew, 2024). We theoretically reframe structural access frictions (such as regulatory boundaries) not as technological barriers, but as possible institutional signals that reshape the

distribution of epistemic trust. This provides theoretical leverage for understanding how global technological hierarchies and domestic legitimacies intersect, offering a fresh perspective on how cognitive inequalities are reproduced in the age of generative artificial intelligence.

Practical Implications

From a governance perspective, the findings highlight that policymakers must recognize trust not merely as a byproduct of technological advancement, but as a crucial psychosocial resource that requires strategic management. In societies characterized by high institutional trust, governments can leverage their symbolic legitimacy to foster early adoption of domestic innovations. However, policymakers need to be vigilant against the risk of cultivating closed epistemic ecosystems. To prevent cognitive echo chambers, regulatory frameworks should balance the promotion of domestic algorithmic security with the facilitation of cross-border interoperability and diverse knowledge integration, understanding that trust management needs governance transparency, not only technical optimization.

For AI developers and industry practitioners, this study underscores that achieving algorithmic competitiveness in localized markets requires far more than mere linguistic translation or interface localization. Developers should prioritize institutional alignment as a component of system design. Building affective trust necessitates the integration of culturally congruent interaction logic and transparent compliance with local ethical norms.

Furthermore, developers of global AI models need to adopt participatory design frameworks that actively address the skepticism of local users, establishing culturally responsive and

institutionally sensitive technological solutions.

Limitations and Future Research

Several limitations warrant consideration for future research. The cross-sectional design inherently restricts the ability to track the temporal evolution of trust formation. Given that generative AI is a rapidly advancing technology and geopolitical environments are highly dynamic, future longitudinal studies are essential to capture how shifting institutional narratives and policy changes iteratively reshape user trust over time.

Additionally, the sample predominantly comprises younger populations who are typically early adopters of AI technologies. While this demographic offers critical insights into emerging digital behaviors, their sociopolitical experiences and interaction frequencies with institutional systems may differ significantly from those of older generations. Future research should strive for broader demographic representation to test the resilience of the Institutional Prism model across various age cohorts and socioeconomic strata.

Moreover, as this study is situated within Chinese societal context, the generalizability of the institutional refraction mechanism requires further empirical validation. Expanding this research to encompass multiple regions, particularly through comparative studies between high institutional trust and low institutional trust settings, would be highly beneficial. Such comparative endeavors could further elucidate how diverse governance paradigms moderate the relationship between AI origin and user trust, ultimately revealing the broader structural forces governing global communication inequalities.

References

- 方兴东 & 宋珂扬. (2025). 全球 AI 治理的科林格里奇困境:多维挑战与协同治理路径. *未来传播*, 32(03), 19-28+116-117.
<https://doi.org/10.13628/j.cnki.zjcmxb.2025.03.003>
- Fang, X., & Song, K. (2025). The Collingridge dilemma in global AI governance: Multidimensional challenges and cooperative governance pathways. *Future Communication*, 32(3), 19–28+116–117.
<https://doi.org/10.13628/j.cnki.zjcmxb.2025.03.003> (In Chinese)
- 刘德寰 & 赵竞鹤. (2023). 中华文化赋能生成式人工智能的本土化发展. *新媒体与社会*, 03, 3-15+386.
- Liu, D., & Zhao, J. (2023). The empowerment of generative AI domesticization by Chinese culture. *New Media and Society (China)*, 3, 3–15+386. (In Chinese)
- 史蒂芬·沃森, 乔纳森·罗米, 文森特·连恩, & 袁莉. (2023). ChatGPT 与社会、教育和技术的纠缠进化——基于系统论视角. *中国教育信息化*, 29(09), 12~20.
- Watson, S., Romi, J., Lian, V., & Yuan, L. (2023). The entangled evolution of ChatGPT, society, education, and technology: A systems theory perspective. *China Educational Informatization*, 29(9), 12–20. (In Chinese)
- 孙昕, 徐志刚, 陶然, & 苏福兵. (2007). 政治信任、社会资本和村民选举参与——基于全国代表性样本调查的实证分析. *社会学研究*, 04, 165-187+245.
<https://doi.org/10.19934/j.cnki.shxyj.2007.04.008>
- Sun, X., Xu, Z., Tao, R., & Su, F. (2007). Political trust, social capital, and villagers' electoral participation: An empirical analysis based on a national representative survey.

- Sociological Studies*, 4, 165–187+245.
- <https://doi.org/10.19934/j.cnki.shxyj.2007.04.008> (In Chinese)
- Amir, S., Luk, S., Saha, S., Tsyrlneva, I., & Teo, M. T. L. (2025). Measuring Social Trust in AI: How Institutions Shape the Usage Intention of AI-Based Technologies. *Human Behavior and Emerging Technologies*. <https://doi.org/10.1155/hbe2/4084384>
- Aoki, N. (2021). The importance of the assurance that “humans are still in the decision loop” for public trust in artificial intelligence: Evidence from an online experiment. *Computers in Human Behavior*, 114, 106572.
- <https://doi.org/10.1016/J.CHB.2020.106572>
- Benbasat, I., & Wang, W. (2005). Trust in and adoption of online recommendation agents. *Journal of the association for information systems*, 6(3), 4.
- Bourdieu, P. (1990). *The logic of practice*. Stanford university press.
- Couldry, N. (2003). Media meta-capital: Extending the range of Bourdieu’s field theory. *Theory and society*, 32(5), 653-677.
- Flew, T. (2024). Mediated trust, the internet and artificial intelligence: Ideas, interests, institutions and futures. *Policy & Internet*, 16(2), 443-457.
- Fosch-Villaronga, E., & Millard, C. (2019). Cloud robotics law and regulation: Challenges in the governance of complex and dynamic cyber–physical ecosystems. *Robotics and autonomous systems*, 119, 77~91.
- Gillespie, N., Lockey, S., Curtis, C., Pool, J., & Akbari, A. (2023). *Trust in Artificial Intelligence: A global study*. <https://doi.org/10.14264/00d3c94>
- Granovetter, M. (1985). Economic action and social structure: The problem of

- embeddedness. *American journal of sociology*, 91(3), 481~510.
- Grimmelikhuijsen, S. (2022). Explaining why the computer says no: algorithmic transparency affects the perceived trustworthiness of automated decision-making. *Public Administration Review*. <https://doi.org/10.1111/puar.13483>
- Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y., De Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human factors*, 53(5), 517~527.
- Hayes, A. F. (2017). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. Guilford publications.
- Hilgers, M., & Mangez, E. (2014). Introduction to Pierre Bourdieu's theory of social fields. In *Bourdieu's theory of social fields* (pp. 1-36). Routledge.
- Horsburgh, S., Goldfinch, S., & Gauld, R. (2011). Is Public Trust in Government Associated With Trust in E-Government? *Social Science Computer Review*, 29, 232–241.
<https://doi.org/10.1177/0894439310368130>
- Kao, Y.-H., & Sapp, S. (2022). The effect of cultural values and institutional trust on public perceptions of government use of network surveillance. *Technology in Society*.
<https://doi.org/10.1016/j.techsoc.2022.102047>
- Kim, K. J., Park, E., Sundar, S. S., & Del Pobil, A. P. (2012). The effects of immersive tendency and need to belong on human-robot interaction. *Proceedings of the seventh annual ACM/IEEE international conference on Human-Robot Interaction*, 207~208.
- Kizilcec, R. F. (2016). How much information? Effects of transparency on trust in an algorithmic interface. *Proceedings of the 2016 CHI conference on human factors in*

computing systems, 2390~2395.

Komiak, S. Y., & Benbasat, I. (2006). The effects of personalization and familiarity on trust and adoption of recommendation agents. *MIS quarterly*, 941~960.

Kostka, G., Steinacker, L., & Meckel, M. (2021). Between security and convenience: Facial recognition technology in the eyes of citizens in China, Germany, the United Kingdom, and the United States. *Public Understanding of Science* (Bristol, England), 30, 671–690. <https://doi.org/10.1177/09636625211001555>

König, P. D., Felfeli, J., Achtziger, A., & Wenzelburger, G. (2022). The importance of effectiveness versus transparency and stakeholder involvement in citizens' perception of public sector algorithms. *Public Management Review* (Vol. 26, pp. 1061–1082). <https://doi.org/10.1080/14719037.2022.2144938>

Lee, Y.-K., Chang, C.-T., Lin, Y., & Cheng, Z.-H. (2014). The dark side of smartphone usage: Psychological traits, compulsive behavior and technostress. *Computers in human behavior*, 31, 373~383.

Li, X. (2025). Public Trust in Global AI Governance Across Geopolitical Rivals. *Global Policy*. <https://doi.org/10.1111/1758-5899.70105>

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of management review*, 20(3), 709~734.

McAllister, D. J. (1995). Affect-and cognition-based trust as foundations for interpersonal cooperation in organizations. *Academy of management journal*, 38(1), 24~59.

McKnight, D. H., Carter, M., Thatcher, J. B., & Clay, P. F. (2011). Trust in a specific technology: An investigation of its components and measures. *ACM Transactions on*

management information systems (TMIS), 2(2), 1-25.

Nass, C., & Moon, Y. (2000). Machines and mindlessness: Social responses to computers.

Journal of social issues, 56(1), 81~103.

Pande, D., Khanal, S., & Taeihagh, A. (2025). Conceptualising Public Trust in High-Risk AI

Policymaking: A Comparative Analysis of Three Asian Cities. *Journal of*

Comparative Policy Analysis: Research and Practice, 27, 578–602.

<https://doi.org/10.1080/13876988.2025.2543546>

Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method

biases in behavioral research: A critical review of the literature and recommended

remedies. *Journal of applied psychology*, 88(5), 879.

Riegelsberger, J., Sasse, M. A., & McCarthy, J. D. (2005). The mechanics of trust: A

framework for research and design. *International journal of human-computer studies*,

62(3), 381~422.

Schiff, K., Schiff, D., Adams, I. T., McCrain, J., & Mourtgos, S. M. (2023). Institutional

Factors Driving Citizen Perceptions of AI in Government: Evidence from a Survey

Experiment on Policing. *Public Administration Review*.

<https://doi.org/10.1111/puar.13754>

Schoorman, F. D., Mayer, R. C., & Davis, J. H. (2007). An integrative model of

organizational trust: Past, present, and future. *Academy of Management Review*, 32(2),

344–354.

Shen, Z., Hu, Y., & Liang, Y. (2026). The Domestic AI Advantage: How the National Identity

of Generative AI Shapes Consumer Adoption of AI -Generated Content. *Journal of*

Consumer Behaviour. <https://doi.org/10.1002/cb.70125>

Song, Y., Natori, T., & Yu, X. (2025). Trusting Humans or Bots? Examining Trust Transfer and Algorithm Aversion in China's E-Government Services. *Administrative Sciences*. <https://doi.org/10.3390/admsci15080308>

Sun, Y., Xu, C., & Xu, H. (2024). Social identity in trusting artificial intelligence agents: Evidence from lab and online experiments. *Managerial and Decision Economics*, 45(8), 5899-5916.

Wang, Y.-F., Chen, Y.-C., Chien, S.-Y., & Wang, P.-J. (2024). Citizens' trust in AI-enabled government systems. *Information Polity*, 29, 293–312. <https://doi.org/10.3233/IP-230065>

Zucker, L. G. (1986). Production of trust: Institutional sources of economic structure, 1840–1920. *Research in organizational behavior*.

Figures and Tables

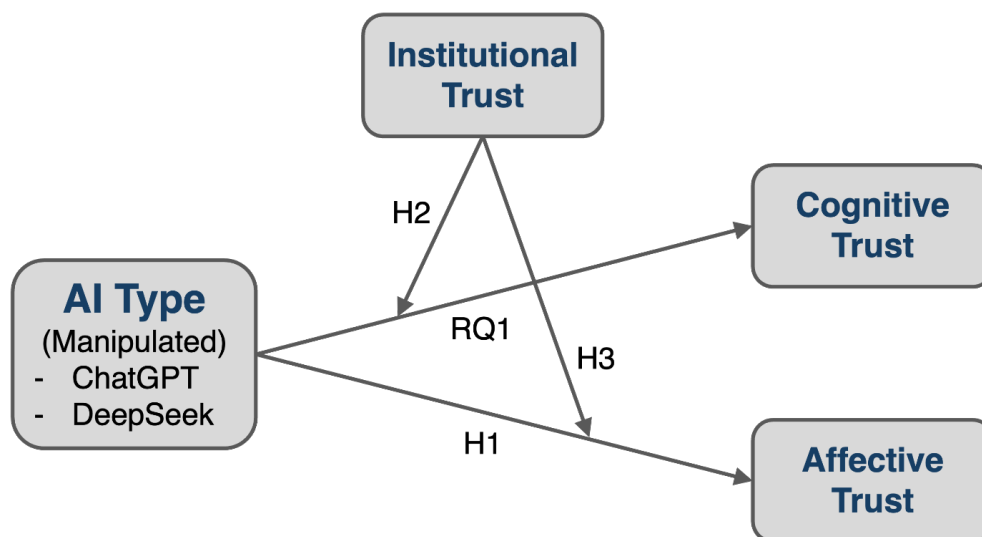
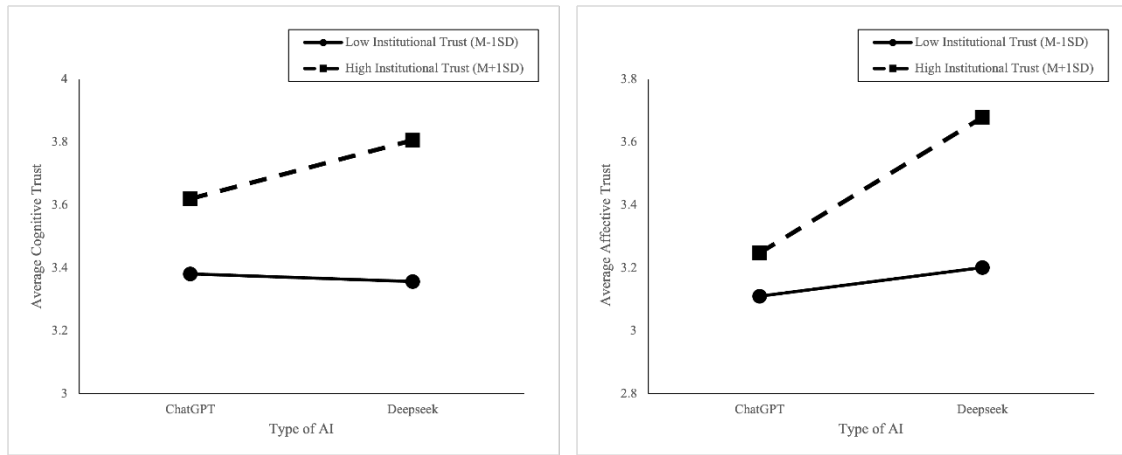


Figure 1 The theoretical model diagram of AI trust formation and institutional moderation.



(a) The moderating effect of cognitive trust

(b) The moderating effect of affective trust

Figure 2 The schematic diagram of the moderating effect

Table 1 Pearson Correlation Matrix of Key Variables

Variable	Mean	SD	AI Type	Affective Trust	Cognitive Trust	Institutional Trust
AI Type	0.500	0.500	1	-	-	-
Affective Trust	3.310	0.693	0.188**	1	-	-
Cognitive Trust	3.541	0.623	0.065	0.568**	1	-
Institutional Trust	3.672	0.769	0.000	0.222**	0.276**	1

**p < 0.01